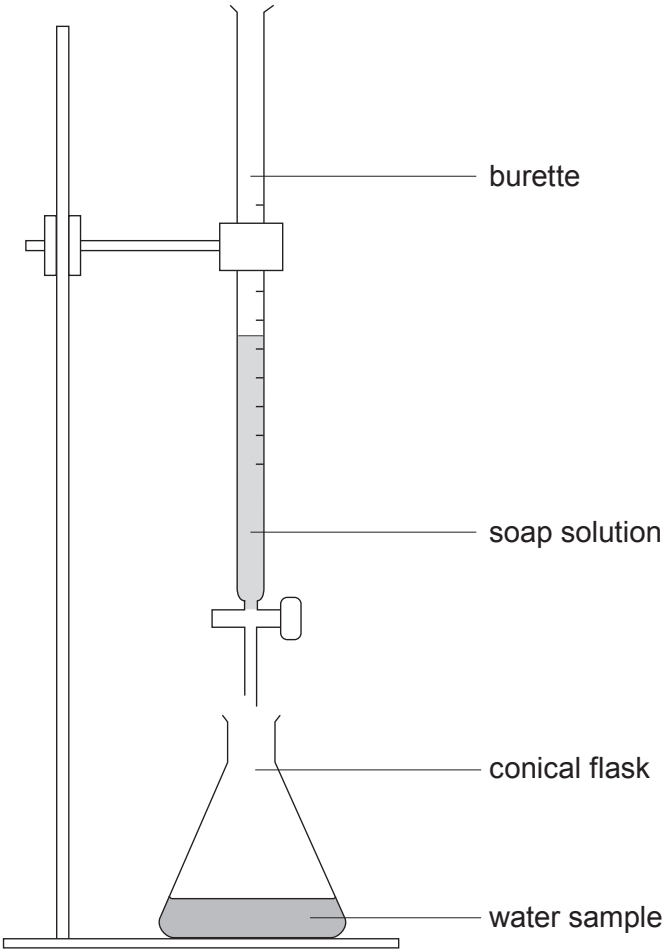


9. Water samples **A**, **B**, **C** and **D** were tested for hardness using the apparatus shown.



Soap solution was added 1 cm³ at a time to each sample and the volume required to produce a permanent lather on shaking was recorded. Each sample was tested before and after boiling. The results are shown in the table.

Water sample	Volume of soap solution required (cm ³)	
	Before boiling	After boiling
A	1	1
B	10	10
C	15	1
D	15	8



Examiner
only

(a) (i) State which water sample contains **only** temporary hardness. Explain your answer. [2]

Water sample

Explanation

(ii) Give **one** similarity in the composition of temporary and permanent hard water. [1]

(b) Discuss the benefits and drawbacks of living in a hard water area. [3]

6

